The remaining block sizes at a critical head

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses: $n = 2r \geq 18$, $G \leq \mathrm{Alt}(\Omega)$ is transitive with connected string C-marking, and every ambient maximal omission is intransitive. The acquired head has support D of size six. Write $E = \Omega \setminus D$, $B = G_{\geq 4}$ and $\sigma = \rho_4$. The group B fixes D pointwise, and all of its marks other than σ fix its anchor $b \in E$.

Theorem 1. *The group B cannot be transitive on E and preserve a block system whose block size a and number z are both at least three.*

Proof. Every distinguished omission in B is intransitive on E: a transitive omission, together with the head and ρ_3 linking it to b, would make the corresponding full ambient omission transitive. Choose a minimal distinguished subset M generating the complete action on the z blocks. Let c count the actual point orbits of $\langle M \rangle$. The acquired actual-orbit forest and Whiston bounds give

$$d - |M| \leq c - 1, \qquad c \leq a, \qquad |M| \leq z - 1, \qquad d = r - 4 = \frac{az}{2} - 1.$$

Thus $(a-2)(z-2) \leq 2$. Since $az = |E| \geq 12$ is even and $a, z \geq 3$, the only possibilities are $(a, z) = (3, 4)$ and $(4, 3)$. Both have $|E| = 12$, $n = 18$, $r = 9$ and $d = 5$. In either case every displayed inequality is an equality: $|M| = z - 1$, $c = a$, and the independent block images generate S_z by Whiston's equality clause. The a actual point orbits each meet every block once. On each such row the block action is faithful, and $\langle M \rangle$ is the diagonal natural S_z on the a rows.

For $(a, z) = (3, 4)$, an element with odd block image acts oddly on each of the three rows and hence oddly on E. It fixes D, contradicting $\langle M \rangle \leq G \leq A_n$.

For $(a, z) = (4, 3)$, the subset M contains σ: every other distinguished mark fixes the block containing b, and cannot generate the transitive block action alone. Its other mark must be ρ_5. Indeed the two nonidentity involutory block images generating S_3 are distinct transpositions and do not commute, whereas σ commutes with every ρ_j for $j \geq 6$. Thus $\langle \sigma, \rho_5 \rangle$ is diagonal natural S_3 on four rows of size three.

The nonidentity even mark ρ_7 centralizes this entire group. The centralizer of natural S_3 on three points is trivial, so a permutation centralizing its diagonal action is exactly a permutation of the four rows, with the unique equivariant bijection between each pair of rows. As ρ_7 fixes b, it fixes the row containing b pointwise. Its remaining action is a nonidentity involution on the other three rows: the identity row permutation would make it fix all of E and also D, contrary to nonidentity. It therefore exchanges two rows, making precisely three point transpositions. This is odd, the final contradiction. □

Attribution and boundary. The critical-head data and full actual-orbit forest/Whiston interfaces are acquired inputs. The new step uses parity in the forced diagonal actions at equality. Neither a projected IP claim nor an alternating rank induction, new finite census or pending packet is used. The cases with block size or number two, primitive B, opposite heads and the unshared endpoint retain their own proof obligations. No whole-original, formal or question credit is claimed.